Database Design and Implementation: E-Commerce System for Video Games and Electronics

Teddy F. Baeny Vargas

University of Niagara Falls

CPSC-500-7:SQL Databases

Dr. Omid Isfahanialamdari

December 9, 2024

Introduction

The primary purpose of any database designed and created for any purpose is to store, data. The one described in this work recreates the operations of an **e-commerce platform** selling a variety of products, including **computers, cell phones, headphones, video games, and accessories**. The **Customers, Employees, Products, Orders, Payments, Shipments and Suppliers** were designed to facilitate the management of the key parts of the business.

In addition to its operational purpose, the database can also serve as a foundation for **analytics and machine learning projects**. With historical data stored in an organized structure, it allows for advanced data analysis, such as customer segmentation, demand forecasting, and product recommendations, all of which contribute to optimizing sales and improving customer satisfaction.

Table Descriptions

Customers Table

This table stores information about the customers who interact with the e-commerce platform. Table 1 shows the attributes and description of each one of them.

Table 1

Customers Table

Attributes	Description
CustomerID (PK)	A unique identifier for each customer.
FirstName	The first name of the customer.
LastName	The last name of the customer.
Email	The customer's email address, used for communication and login purposes.
Phone	The customer's phone number.
Address	The residential or business address of the customer.
Country	The country where the customer resides.
City	The city where the customer resides.
CreatedDate	The timestamp when the customer record was created in the system.
UpdatedDate	The timestamp when the customer record was last updated.

Products Table

This table stores product details, including name, description, price, and stock quantity, and links each product to its brand, category, and platform. Table 2 shows the attributes and description of each one of them.

Table 2

Products Table

Attributes	Description
ProductID (PK)	A unique identifier for each product.
ProductName	The name of the product (e.g., Gaming Laptop, Wireless Mouse).
ProductDescription	A description of the product, detailing its features.
Price	The price of the product.
StockQuantity	The number of units available in stock for the product.
BrandID (FK to Brands)	A reference to the brand of the product.
CategoryID (FK to Categories)	A reference to the category the product belongs to.
PlatformID (FK to Platforms)	A reference to the platform that the product is related to (e.g., Windows, PlayStation).
CreatedDate	The timestamp when the product was added to the system.
UpdatedDate	The timestamp when the product record was last updated.

Orders Table

This table stores order details, including customer information, order status, and total amount. Table 3 shows the attributes and description of each one of them.

Table 3

Orders Table

Attributes	Description
OrderID (PK)	A unique identifier for each order.
EmployeeID (FK to Employees)	The ID of the employee who processed the order.
CustomerID (FK to Customers)	A reference to the customer who placed the order.
OrderDate	The date when the order was placed.
OrderStatus	The status of the order (e.g., Pending, Completed, Cancelled).
TotalAmount	The total value of the order.
CreatedDate	The timestamp when the order was created in the system.
UpdatedDate	The timestamp when the order record was last updated.

OrderDetails Table

This table stores the details of the products within an order, including quantity and price.

Table 4 shows the attributes and description of each one of them.

Table 4

OrderDetails Table

Attributes	Description
OrderID (FK to Orders)	A unique number identifying every order.
ProductID (FK to Products)	A number that identifies the product reviewed by the customer.
Quantity	Number used to know the quantity of goods the customer purchased.
UnitPrice	The price of the article at the time the customer put an order.
CreatedDate	The timestamp when the order was created in the system.
UpdatedDate	The timestamp when the order record was last updated.

Payments Table

This table tracks payments made for orders, including payment method and amount. Table 5 shows the attributes and description of each one of them.

Table 5

Payments Table

Attributes	Description
PaymentID (PK)	A number identifying each payment by the customers.
OrderID (FK to Orders)	A number that identifies the order in a unique way
PaymentDate	The date when the payment was made by the customer.
PaymentAmount	A number that represents the total amount of money paid by the customer.
PaymentMethod	Text that describes the method used to pay (e.g., Credit Card, PayPal, Bank Transfer).
CreatedDate	The date when the payment record was created in the system.
UpdatedDate	The timestamp when the payment record was last updated.

Promotions Table

The table is used for storing all the promotions active or not, including details such as the discount percentage, start and end dates. Table 6 shows the attributes and description of each one of them.

Reviews Table

This table stores the customers' comments for every product they have purchased and hag something to say about it. Table 7 shows the attributes and description of each one of them.

Table 6

Promotions Table

Attributes	Description
PromotionID (PK)	A unique identifier for each promotion.
Name	The name of the promotion (e.g., Summer Sale, Black Friday Deal).
Description	A detailed description of the promotion, explaining the terms and conditions.
DiscountPercentage	The discount offered in the promotion as a percentage.
StartDate	The date when the promotion starts.
EndDate	The date when the promotion ends.
IsActive	A flag indicating whether the promotion is currently active (1 = Active, 0 = Inactive).
CreatedDate	The timestamp when the promotion was created in the system.
UpdatedDate	The timestamp when the promotion record was last updated.

Table 7

Reviews Table

Attributes	Description
ReviewID (PK)	A number identifying each review in a unique way
CustomerID (FK to Customers)	The identification of the customer who wrote the review.
ProductID (FK to Products)	A number that identifies the product reviewed by the customer.
Rating	A number between 1 and 5, associated with the feeling the user has about the product purchased.
Comment	The text typed by the user.
ReviewDate	The date when the review was created by the user.

Shipments Table

Tracks shipment details for orders. Table 8 shows the attributes and description of each one of them.

Table 8

Shipments Table

Attributes	Description
ShipmentID (PK)	Unique number identifying the shipment
OrderID (FK to Orders)	A reference to the order associated with the shipment.
EmployeeID (FK to Employees)	The identification of the employee in charge of the shipment.
ShipmentDate	The date when the sold goods were sent to the customer.
TrackingNumber	The alphanumerical number provided by the shipping carrier.
ShipmentStatus	The status of the shipment (e.g., Pending, Shipped, Delivered).
CreatedDate	The date when the shipment record was created in the system.

Suppliers Table

Stores details about suppliers providing products. Table 9 shows the attributes and description of each one of them.

Table 9

Suppliers Table

Attributes	Description
SupplierID (PK)	A unique identifier for each supplier.
SupplierName	The name of the supplier (e.g., Tech Supplies Inc.).
ContactName	The name of the primary contact person for the supplier.
Phone	The phone number of the supplier.
Email	The email address of the supplier.
Address	The address of the supplier.
CreatedDate	The timestamp when the supplier record was created in the system.
UpdatedDate	The timestamp when the supplier record was last updated.

Supplier_Products Table

Associates products with their suppliers. Table 10 shows the attributes and description of each one of them.

Table 10

Suppliers_Products Table

Attributes	Description
SupplierID (FK to Suppliers)	A reference to the supplier who provides the product.
ProductID (FK to Products)	A reference to the product provided by the supplier.

Promotions_Products Table

Links promotions to applicable products. Table 11 shows the attributes and description of each one of them.

Table 11

Promotions_Products Table

Attributes	Description
PromotionID (FK to Promotions)	A reference to the promotion applied to the product.
ProductID (FK to Products)	A reference to the product that is part of the promotion.

Platforms Table

Stores information about platforms associated with products. Table 12 shows the attributes and description of each one of them.

Table 12

Platforms Table

Attributes	Description
PlatformID (PK)	A unique identifier for each platform.
PlatformName	The name of the platform (e.g., Windows, PlayStation).
PlatformDescription	A description of the platform, explaining its purpose or features.
CreatedDate	The timestamp when the platform record was created in the system.
UpdatedDate	The timestamp when the platform record was last updated.

Brands Table

Stores information about product brands. Table 13 shows the attributes and description of each one of them.

Table 13

Brands Table

Attributes	Description
BrandID (PK)	A unique identifier for each brand.
BrandName	The name of the brand (e.g., Dell, Sony).
BrandDescription	A description of the brand, explaining its background or notable products.
CreatedDate	The timestamp when the brand record was created in the system.
UpdatedDate	The timestamp when the brand record was last updated.

Categories Table

Manages product categories. Table 14 shows the attributes and description of each one of them.

Table 14

Categories Table

Attributes	Description
CategoryID (PK)	A unique identifier for each category.
CategoryName	The name of the category (e.g., Laptops, Accessories).
CategoryDescription	A description of the category, explaining its purpose or the types of products it includes.
CreatedDate	The timestamp when the category record was created in the system.
UpdatedDate	The timestamp when the category record was last updated.

Employees Table

Stores details about employees and their roles. Table 15 shows the attributes and description of each one of them.

Table 15

Employees Table

Attributes	Description
EmployeeID (PK)	A unique identifier for each employee.
Name	The name of the employee.
DepartmentID (FK to Departments)	A reference to the department the employee works in.
Position	The job title or position of the employee.
Salary	The salary of the employee.
HireDate	The date the employee was hired.
CreatedDate	The timestamp when the employee record was created in the system.
UpdatedDate	The timestamp when the employee record was last updated.

Departments Table

Stores details about departments. Table 16 shows the attributes and description of each one of them.

Table 16

Departments Table

Attributes	Description
DepartmentID (PK)	A unique identifier for each department.
DepartmentName	The name of the department (e.g., Sales, Marketing).
DepartmentDescription	A description of the department's function.
CreatedDate	The timestamp when the department record was created in the system.
UpdatedDate	The timestamp when the department record was last updated.

Relationships Between Tables

Customers and Orders

A customer can make zero, one, or many orders over time and each order can only be associated with exactly one customer. Thus, the relationship is one-to-many.

Orders and OrderDetails

It is a one-to-many relationship. Each order can have one or many products, represented by multiple records in the the **OrderDetails** table while each order detail belongs to a unique specific order.

Products and OrderDetails

It is a one-to-many relationship. A single product can be associated in zero, one, or many orderDetails, but every orderDetail is associated with one and only one product.

Products and Reviews

It is also a one-to-many relationship. A product can have zero, one, or many customer reviews, with each review linked to exactly one product.

Customers and Reviews

It is a one-to-many relationship. A single customer can write zero, one, or many reviews for different products, but each review is written by exactly one customer.

Orders and Payments

It is a one-to-one relationship. Each order is associated with exactly one payment, and every payment corresponds to one order only.

Products and Promotions_Products

The **Products** and **Promotions** tables have a many-to-many relationship. To achieve that relationship, it is necessary the junction **Promotions_Products** table. A single product can be included in zero, one, or many promotions, and a single promotion can apply to zero, one, or many products.

Suppliers and Supplier_Products

The **Suppliers** and the **Products** tables are in a many-to-many relationship and use the **Supplier_Products** table as the junction between them. A supplier can provide zero, one, or many products, and a product can be supplied by zero, one, or many suppliers.

Products and Brands, Products and Categories and Products and Platforms

It is a one-to-many relationships with the **Brands**, **Categories**, and **Platforms** tables. Each product is associated with exactly one brand, one category, and one platform, while each brand, category, or platform can include zero, one, or many products.

Employees and Orders

It is a one-to-many relationship where an employee can process zero, one, or many orders, and each order is processed by exactly one employee.

Employees and Shipments

It is a one-to-many relationship. An employee can manage multiple shipments, but each shipment is handled by a single employee.

Employees and Departments

It is a one-to-many relationship. Each employee belongs to exactly one department, while a department can include zero (the case of a new department), one, or many employees.

Data Types and Constraints

Data Types

The **Integer (INT)** data type is used to store whole numbers without any decimal points.

The **Variable Character (VARCHAR)** data type was used for text attributes.

The **Decimal (DECIMAL)** data type was used for attributes requiring precision and are used for monetary values like the attributes Price, TotalAmount, and PaymentAmount.

The **Date (DATE)** data type was used for attributes that store calendar dates, such as OrderDate, PaymentDate, and RegistrationDate.

The **Timestamp (TIMESTAMP)** data type was used in the attributes that record the exact date and time. The property called DEFAULT CURRENT_TIMESTAMP ensures that these fields are automatically populated with the timestamp during insertion or updates.

The **Boolean (BOOLEAN)** data type was used in the binary values present in the database, such as the attribute IsActive in the **Promotions** table.

Constraints

The **Primary Key (PK)** constraint makes each record in the table has a unique identifier. The property AUTO_INCREMENT automatically generate unique values for new records.

The **Foreign Key (FK)** constraint links tables through related fields.

The **Unique Constraint** specify that the columns need to be distinct values across all records.

The **Check Constraint** was used to secure that the values to be inserted in a column meet specific conditions.

The **Not Null Constraint** requires that the column to be inserted does not have empty values.

Data Normalization

The structured approach of organizing data to minimize redundancy and ensure data integrity is called database normalization. This concept, introduced by E.F. Codd in 1970 as part of the relational database framework, remains a fundamental aspect of database design. Normalization occurs in stages called normal forms, with each stage addressing specific types of data irregularities to refine the database structure.

First Normal Form (1NF)

Remove repeating groups from individual tables.

Establish separate tables for each related data set.

Use a primary key to uniquely identify each set of related data.

All the tables designed and created for the schema comply with these restrictions. Figure 1 shows the ERD

Second Normal Form (2NF)

The tables must satisfy 1NF

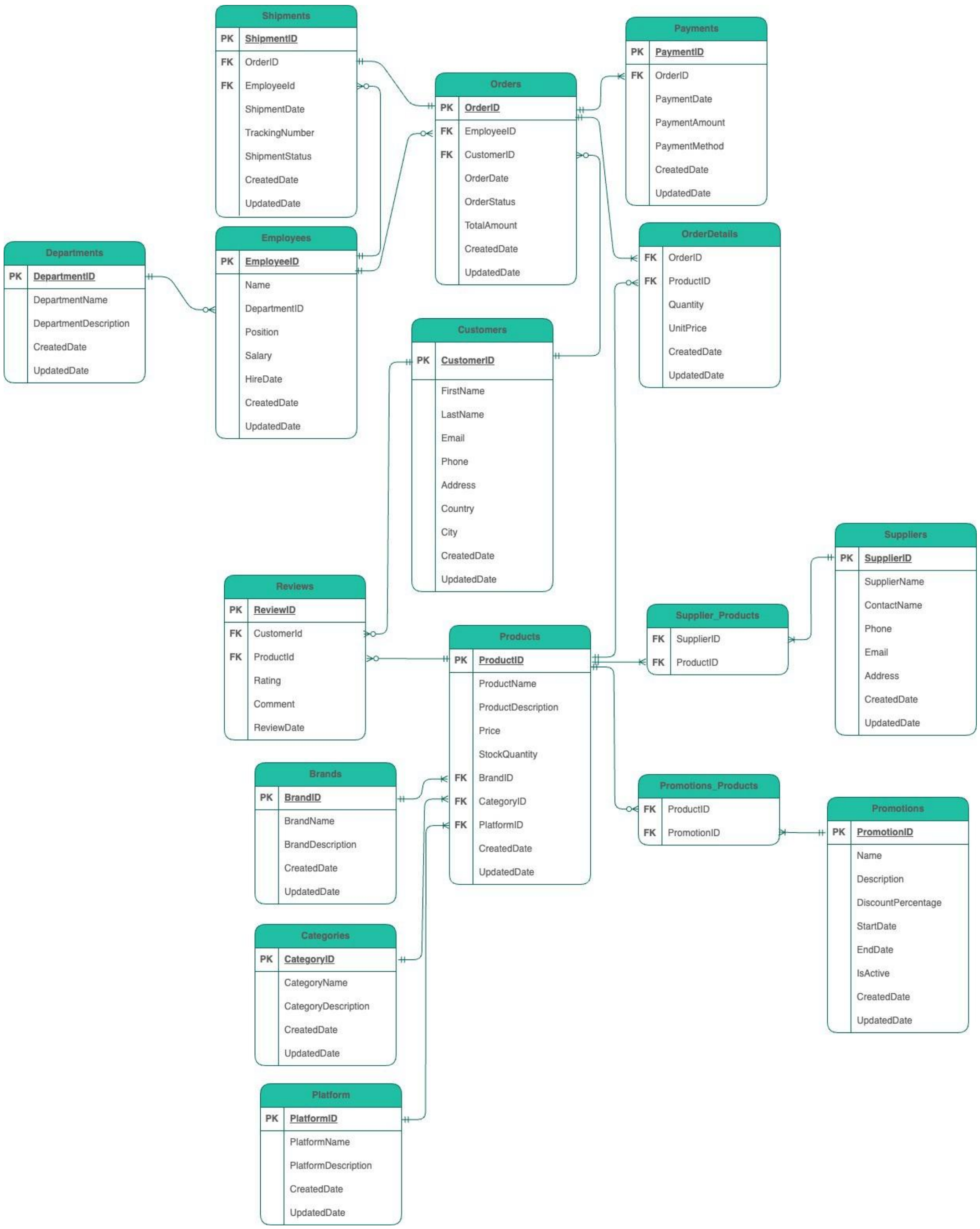
Create distinct tables for sets of values that are relevant to multiple records.

Connect these tables using a foreign key.

In the design, there is not a single attribute that depends only on part of the key, as there's a single primary key. The database complies with the Second Normal Form.

Figure 1

ERD for the database



Note. The ERD also can be found in this link:

https://drive.google.com/file/d/1Th2QyOVMRH0KePPt6MZwY62pR1TcpM5u/view?usp=sharing

Third Normal Form (3NF)

The tables must satisfy 2NF

Remove any field that does not depend on the key. There shouldn't be a transitive dependency (non-key attributes must not depend on other non-key attributes). The tables also comply with these restrictions. Figure 1 show the complete ERD with all the tables in the final design implementation complying with 1NF, 2NF, 3NF.

SQL Indexes

An index reduces disk input/output operations by providing a quick access path to locate data efficiently. While indexes improve the performance of SELECT queries and WHERE clauses, they can slow down operations like INSERT and UPDATE due to the additional overhead of maintaining the index. Creating or removing an index does not alter the actual data in the table.

Indexes Used

For the database, a couple of indexes were created in order to improve the SELECT and WHERE queries in the database. Figure 2 show the index created for the **Categories** table. Recurrent queries using the name (e.g., *SELECT * FROM Categories WHERE CategoryName = 'Electronics'*) will significantly improve the query performance if an index is used.

Figure 2

Index created for the Categories Table

```
CREATE INDEX idx_categories_categoryname ON Categories (CategoryName);
```

Data Retrieval Statements (DQL)

After the data was ready, Data Query Language (DQL) techniques were used. These inquiries are designed to satisfy the needs of small businesses, assisting the team in time optimization, problem solving, and rapid information retrieval. The database facilitates improved decision-making by providing insightful information, which boosts sales and streamlines operations.

View

This view provides the planning team with a comprehensive overview of sales, inventory, and promotions. By offering a clear picture of current operations, it helps the team prepare effectively for upcoming seasonal sales, ensuring they can anticipate demand, optimize inventory, and align promotions to meet business goals.

Window Function

This analysis provides important insights into employee remuneration by taking a close look at the salary distribution across all divisions. The report deconstructs the following important KPIs for every department:

Average Salary: The department's average salary, which provides insight into general trends in compensation.

Maximum compensation: The department's maximum compensation, which identifies the highest earners.

Minimum Salary: The department's lowest salary, which may indicate entry-level pay or possible anomalies.

Additionally, the report enables significant departmental and organizational comparisons. Within departments, a clear picture of each worker's performance in relation to their peers is provided by comparing their salaries. Businesses can identify high-performing workers, better understand salary distribution, and make more strategic decisions about workforce planning, salary changes, and promotions by incorporating these data. Figure 3 has the total query.

Figure 3

Query with Data Retrieval Statements

```
SELECT
    E.EmployeeID,
    E.Name AS EmployeeName,
    D.DepartmentName,
    E.Salary,
    ROUND(AVG(E.Salary) OVER (PARTITION BY D.DepartmentID), 2) AS AverageSalaryInDepartment,
    ROUND(MAX(E.Salary) OVER (PARTITION BY D.DepartmentID), 2) AS MaxSalaryInDepartment,
    ROUND(MIN(E.Salary) OVER (PARTITION BY D.DepartmentID), 2) AS MinSalaryInDepartment,
    RANK() OVER (ORDER BY E.Salary DESC) AS OverallSalaryRank,
    ROW_NUMBER() OVER (PARTITION BY D.DepartmentID ORDER BY E.Salary DESC) AS SalaryRankInDepartment
FROM
    Employees E
    INNER JOIN Departments D ON E.DepartmentID = D.DepartmentID
ORDER BY
    D.DepartmentName, Salary DESC;
```

CTE

This search examines consumer purchases, including overall expenditure, patterns in payment methods, and product evaluations. For deeper insights, it integrates product ratings and payment details with the total spending per client, which is determined using a Common Table Expression (CTE). This approach ensures a comprehensive analysis by connecting financial, behavioral, and feedback data. It supports more informed decision-making, enabling businesses to enhance customer satisfaction, improve product offerings, and optimize sales strategies. Figure 4 has a query a CTE created.

Figure 4

Query with CTE

```sql
WITH CustomerSpending AS (
    SELECT
        C.CustomerID,
        CONCAT(C.FirstName, ' ', C.LastName) AS CustomerName,
        SUM(O.TotalAmount) AS TotalSpent
    FROM
        Customers C
        INNER JOIN Orders O ON C.CustomerID = O.CustomerID
    GROUP BY
        C.CustomerID, C.FirstName, C.LastName
)

SELECT
    CS.CustomerID,
    CS.CustomerName,
    CS.TotalSpent,
    COUNT(DISTINCT O.OrderID) AS TotalOrders,
    MAX(P.PaymentMethod) AS MostRecentPaymentMethod,
    COUNT(R.ReviewID) AS TotalReviews,
    COALESCE(ROUND(AVG(R.Rating), 2),0) AS AverageRating
FROM
    CustomerSpending CS
    LEFT JOIN Orders O ON CS.CustomerID = O.CustomerID
    LEFT JOIN Payments P ON O.OrderID = P.OrderID
    LEFT JOIN Reviews R ON O.CustomerID = R.CustomerID
GROUP BY
    CS.CustomerID, CS.CustomerName, CS.TotalSpent
ORDER BY
    CS.TotalSpent DESC;
```

Query

This query integrates data from suppliers, products, promotions, and reviews to deliver a comprehensive overview of key business metrics, including supplier performance, product inventory levels, and customer satisfaction.

Measuring the success of promotions and their influence on consumer perception is also helpful, as is identifying high-performing suppliers, addressing underperformance, and optimizing inventory management to satisfy customer demand without overstocking.

Businesses have a comprehensive perspective that aids in strategic decision-making for supplier relationships, inventory optimization, and customer experience enhancement by combining these vital data points.

Figure 5

Example of Query

```sql
SELECT
    S.SupplierID,
    S.SupplierName,
    COUNT(DISTINCT SP.ProductID) AS TotalProductsSupplied,
    SUM(P.StockQuantity) AS TotalStockAvailable,
    ROUND(AVG(R.Rating), 2) AS AverageProductRating,
    COUNT(DISTINCT PR.PromotionID) AS PromotionsActive
FROM
    Suppliers S
    INNER JOIN Supplier_Products SP ON S.SupplierID = SP.SupplierID
    INNER JOIN Products P ON SP.ProductID = P.ProductID
    LEFT JOIN Reviews R ON P.ProductID = R.ProductID
    LEFT JOIN Promotions_Products PP ON P.ProductID = PP.ProductID
    LEFT JOIN Promotions PR ON PP.PromotionID = PR.PromotionID AND PR.IsActive = TRUE
GROUP BY
    S.SupplierID, S.SupplierName
ORDER BY
    TotalProductsSupplied DESC, AverageProductRating DESC;
```

To retrieve orders with a pending shipping status, figure 6, this query aggregates information from Orders, Shipments, Customers, and Employees. This aids the team in setting priorities and carrying out required tasks.

Figure 6

Example of Query orders with a pending shipping status

```sql
SELECT
    O.OrderID,
    CONCAT(C.FirstName, ' ', C.LastName) AS CustomerName,
    C.Email AS CustomerEmail,
    E.Name AS AssignedEmployee,
    S.ShipmentStatus,
    O.OrderStatus,
    S.ShipmentDate,
    O.OrderDate,
    DATEDIFF(CURDATE(), O.OrderDate) AS DaysSinceOrder
FROM
    Orders O
    LEFT JOIN Shipments S ON O.OrderID = S.OrderID
    INNER JOIN Customers C ON O.CustomerID = C.CustomerID
    LEFT JOIN Employees E ON S.EmployeeID = E.EmployeeID
WHERE
    O.OrderStatus = 'Pending'
    OR S.ShipmentStatus IS NULL -- in case shipment record do not exists yet
ORDER BY
    O.OrderDate ASC, DaysSinceOrder DESC;
```

Use Cases for Machine Learning

Customer Segmentation

The **Customers**, **Orders**, and **OrderDetails** tables are crucial for this analysis. Aggregating the total spending, the frequency of purchases, and product preferences allows clustering algorithms (e.g., K-Means) to define these segments.

Sales Forecasting

By using sales data from the **Orders** and **OrderDetails** tables using time intervals, the businesses can use time-series models like ARIMA for accurate predictions.

Product Recommendations

Extracting data from the **OrderDetails**, **Products** and **Orders** tables and also using filtering can identify the products frequently purchased together increasing cross-selling opportunities.

Sentiment Analysis on Reviews

The comments extracted from the **Reviews** table can be processed to apply them to NLP models to classify the customers feelings as positive, negative, or neutral helping to identify common complaints or areas for improvement.

Inventory Optimization

The **OrderDetails** and the **Products** tables contain data that predictive models can be use in to forecast demand for each product helping to maintain optimal stock levels and improve supplier coordination.

Promotion Effectiveness

The **Promotions**, **Promotions_Products**, and **Orders** tables and the data that they contain can help to measure the impact of promotions on sales. Sales data during promotional periods could be analyzed and compared to non-promotional periods in order to identify the most effective promotions.

Churn Prediction

The **Customers** and **Orders** tables are crucial for this analysis. With the customer behavior, such as the time a customer made the last purchase and total lifetime spending, machine learning models can be trained to predict the likelihood of churn.